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ORIGINAL ARTICLE

A trial of risk-adapted prostate cancer screening in a federally supported health center network serving a high-risk population

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Abstract

Background: Although Black men experience earlier onset and higher mortality from prostate cancer (PCa), no studies to date have tested interventions that encourage adoption of risk-adapted prostate-specific antigen (PSA) screening. The authors conducted a 15-month trial in clinics serving Chicago communities with high PCa mortality and Black population density. Their objective was to determine the impact of "Smart PSA" guidelines on screening behavior and biopsy outcomes.

Methods: Through presentations, in-clinic reminders, and on-site visits, primary care providers (PCPs) were exposed to basic PSA guidelines recommending starting age, repeat test interval, and stopping age, based on race, family history, and baseline PSA. For the intervention period and a matched control period 3 years earlier, the authors extracted data from clinical databases on all visits by men age 35 and over, at six clinics. The primary end point was change in incidence of a screening PSA linked to a PCP encounter.

Results: A total of 4252 and 4022 patients, similar in age and race, had qualifying encounters during intervention and pre-intervention, respectively. PSA screening following encounters increased from 18.7 to 33.0 per 100 patients ($\Delta = 14.3$; 95% CI, 12.4–16.1) with odds ratio = 2.43 (95% CI, 2.17–2.71), adjusted for pre-specified covariates. Guideline compliance was indicated by a larger increase in screening of younger Black patients, and earlier repeat testing after a borderline PSA. Increased screening of men 70 and older indicated potential noncompliance. Biopsy incidence increased from 0.57 to 0.92 per 100 ($\Delta = 0.35$; 95% CI, −0.03 to 0.75), with PCa prevalence at biopsy increasing from 43% to 74%. Total PCa incidence rose 2.7-fold (0.25 to 0.68 per 100, $p = .007$), with incidence of cancer beyond grade group 1 increasing 3-fold ($p = .008$).

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Conclusion: These findings demonstrate that risk-adapted PSA guidelines can be adopted and sustained by PCPs in high-risk settings. Although PCa incidence increased substantially, the predominance of aggressive cancer was unchanged, consistent with the effect of intensified screening in a high-risk population with sparse prior screening for a cancer with long latency. Sustained intensification should eventually cause a new equilibrium, necessitating additional efforts to avoid overdiagnosis.

Trial Registration: The trial registration is ClinicalTrials.gov NCT04782713

KEYWORDS
clinical trial, nonrandomized, populations at risk, primary care, prostate cancer, screening

INTRODUCTION

Screening for prostate cancer (PCa) with blood levels of prostate-specific antigen (PSA) has had a turbulent history since it was introduced approximately 35 years ago. Its rapid adoption and relatively long lead time[1] produced an unprecedented surge in PCa incidence in the United States, triggering legitimate concerns about overdiagnosis and overtreatment, even as mortality rates were falling significantly.[2] In response, the influential US Preventive Services Task Force (USPSTF) actively discouraged PSA screening in 2012 before shifting to support for individualized decisions.[3] Meanwhile, the racial disparity in mortality from prostate cancer, the largest among all major cancers, has persisted to the present day. This disparity has multiple causes, including substandard care and biological factors yet to be fully defined, but the greatest remediable factor at present is inadequate early detection.[4–6]

In areas of the United States with large Black populations, late diagnosis is common. In Chicago, for example, some neighborhoods with a high density of Black residents have PCa mortality rates approximately three times higher than national age-adjusted rates and potentially aggressive cancer at diagnosis is the norm. Concurrently, many primary care providers (PCPs), who are largely responsible for early detection of cancer, are uncertain about the value of PSA screening and the best way to conduct it.[7–9] Screening protocols tailored to a patient's risk level are standard for other major cancers because this is known to improve the balance between risk and benefit. However, authoritative bodies have been cautious about supporting risk-adapted PCa screening, in part due to a lack of empirical data to support such protocols and have only recently begun to promote tailored guidelines.[10–12] Age, race, and family history are long-established risk factors for PCa; but consistent findings from epidemiological studies over the past decade have added another important one: having an elevated PSA within the normal range at midlife.[13]

We used the best available evidence to design risk-adapted ("Smart") PSA guidelines that could accommodate the busy schedules of PCPs who typically see patients with complex medical and social issues. In a federally qualified health center (FQHC) network serving a high-risk population, we conducted a single-arm trial with historical controls to determine the impact of a PCP-focused intervention encouraging implementation of these guidelines, on the primary end point—likelihood of a patient receiving a screening PSA order after a PCP encounter—and secondary end points focused on PSA levels and prostate biopsy outcomes.

MATERIALS AND METHODS

Overall design and study population

The direct recipients of the intervention were PCPs and clinic staff; data from electronic health records and laboratory databases were used to measure the intervention's impact. The study population comprised men aged 35 years or more with no history of PCa who were outpatients at six neighborhood clinics operated by the Mile Square Health Center (MSHC) in Chicago. MSHC is part of the University of Illinois at Chicago (UIC) and is the only federally qualified health center (FQHC) owned and operated by an academic medical center. All six sites are located in areas that are medically under-served, with predominant minority populations. MSHC sites are staffed by family physicians and nurse practitioners affiliated with the University of Illinois Health System; the majority do not hold faculty appointments. In Figure 1A, site locations are superimposed on maps of PCa mortality and Black population density by community area.

The intervention extended from November 11, 2020 through February 28, 2022, and the pre-intervention from matching dates in 2017 through 2019 to control for seasonal variation. All study data were obtained from electronic health records and laboratory databases. The protocol was approved by the Research Council and Community Board of MSHC and by the UIC institutional review board. The study is registered with ClinicalTrials.gov as NCT04782713.

The Smart PSA intervention

The Prostate Cancer Working Group within the University of Illinois Cancer Center created a multi-disciplinary committee (led by

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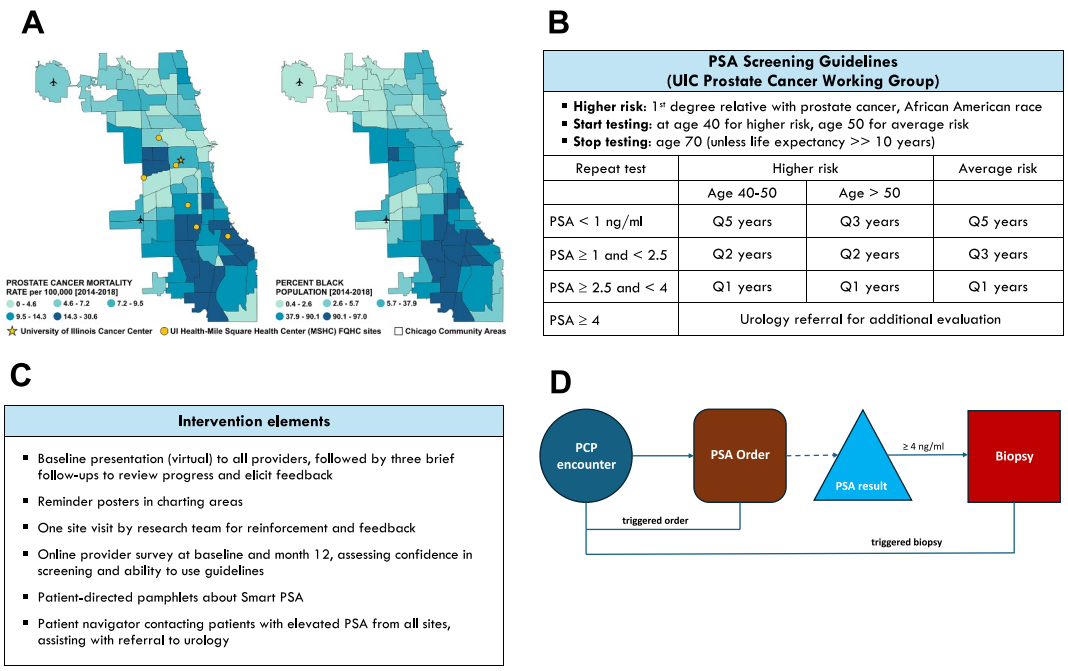


FIGURE 1 (A) Community area (neighborhood) maps of Chicago showing age-adjusted prostate cancer mortality and location of the trial sites (left), and density of Black population (right). (B) Risk-adapted screening guidelines used during intervention. (C) Intervention elements and (D) schematic showing analyzed PSA orders and biopsies triggered by primary care provider encounters during the intervention or pre-intervention period.

coauthors P.H.G. and M.R.A.) to devise guidelines for risk-adapted PSA screening. The committee included expertise in cancer epidemiology, urologic oncology, and primary care and its goal was to produce guidelines, feasible within primary care practice, that address four questions: what age to start screening, how often to re test, what age to stop, and what PSA threshold to use for urology referral, while accounting for race, family history, and PSA level. Routine digital rectal examination was not recommended. Figure 1B shows the screening guidelines and Figure 1C shows a summary of each intervention element. A schematic in Figure 1D shows how PSA orders and subsequent biopsies were linked to a specific visit. Images of the patient brochure and in-clinic reminder poster are shown in Figure S1. PCPs were surveyed at baseline and 12 months later, about their knowledge concerning the benefits and risks of PSA screening, their confidence in shared decision-making, and the feasibility of implementing the guidelines. Patient navigators were assigned to contact patients with PSA >4 to facilitate the referral to urology. Surveys and post-result navigation were considered part of the intervention because both could affect awareness of the guidelines.

Details on cohort eligibility and acquisition of data on patients, visits, PSA orders, and biopsy outcomes are provided in the Supporting Information.

Data analysis

Demographic and clinical characteristics of patients in the cohorts were compared using the appropriate tests for either categorical or continuous variables. The primary end point was evaluated by comparing the difference in the cumulative incidence of PSA screening between study periods, expressed as the number of screening events per 100 eligible patients. A screening event, subsequently referred to as a "triggered PSA order" was defined as an order placed for a screening PSA on the same day as an encounter with an eligible PCP or any day before a subsequent visit within the study interval. Screening orders versus clinically indicated orders were distinguished using internal billing codes specific for PSA screening and total serum PSA, and by censoring all visits following a PSA <0.1, which would indicate prior prostatectomy.

We used multivariable logistic regression, adjusting for six pre-specified variables: age, race, total visits, prior PSA, prior prostate biopsy, and visit site, to measure the independent relative odds of screening during intervention. A seventh pre-specified covariate, PCP type, was dropped due to excessive missing data. Because 1541 patients were members of both cohorts, we evaluated separate models that included all subjects, unique subjects only, and repeat subjects only. For repeat subjects we used conditional logistic regression to account for paired data. Model coefficients were nearly identical for all three models, indicating that bias from within-person correlation was negligible. Therefore, we report results from the all-subjects model. Because the cohorts were consecutive in time, we tested various time windows before baseline for prior PSA and biopsy when assessing balance between the cohorts but used no windows for these covariates when measuring intervention effect.

Effect modification was investigated using stratified comparison of the difference in screening incidence across various subgroups,

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with calculation of 95% confidence intervals.[14] Although absolute change in screening was the main end point, we also assessed effect modification on a relative scale by adding multiplicative interaction terms to the basic model. PSA levels were compared by χ^2 across the <4, 4–10, and >10 ng/mL thresholds and by *t*-test on log-transformed concentrations. Biopsy outcomes were compared using the Fisher exact test for grade group and modified National Comprehensive Cancer Network (NCCN) risk level, which excluded clinical stage and gland volume due to data availability. All statistical tests were two-sided. Statistical analyses were conducted using SAS (version 9.4). Data processing and visualization were performed using R (version 4.3.2).

RESULTS

Table 1 shows characteristics of the pre-Intervention and intervention cohorts, which were similar in terms of patient numbers, patient ages, percentage of Black patients, and number of PCPs. The number of visits per patient was also quite similar, although the percentage of patients with the highest number of visits declined slightly. There were some significant changes in the number of visits per site. PSA testing and prostate biopsy before baseline were comparable between time periods.

The likelihood of receiving PSA screening increased during intervention by 76%, from 18.7 to 33.0 per 100 patients (Δ = 14.3; 95% CI, 12.4–16.1) (Figure 2A). Black patients overall were more likely to be screened during both periods; moreover, in the important 60- to 69-year age group, Black men surpassed non-Black men after lagging behind during pre-intervention. Screening at age 70-plus increased markedly in both race groups, although relatively few patients were involved. Triggered PSA orders rose rapidly over the first 4 months of intervention and remained steady until a drop-off in the final month (Figure 2B). In comparison, encounters linked to orders varied little during pre-intervention.

Multivariable models showed that the screening increases were not attributable to several potential confounding factors specified a priori. In a base model without interactions, the adjusted odds ratio for intervention was 2.43 (95% CI, 2.17–2.71) (Table S1). The anticipated associations between age and screening were observed, except for the 70- to 75-year-olds, whose odds of screening were indistinguishable from the 50- to 59-year-old referent. Independent odds ratios were significantly elevated for Black patients and those with more than one visit or pre-baseline PSA >4, and varied by clinical site. Extending the model to include interactions revealed modifications of the intervention's impact on a relative scale. Predicted probabilities of a PSA order from the extended model illustrate the significant variation in response among patients with different number of visits and clinical site (Figure S2).

Stratified analyses revealed differences in the intervention's absolute impact on specific patient subgroups (Table 2). Some differences indicate that guideline compliance occurred; for instance, screening incidence for men under age 50 increased by 14.5 per 100 for Black men versus 8.5 per 100 for non-Black men (p = .021). Furthermore, the guidelines recommended a repeat PSA in 1 year for all patients with PSA = 2.5–4.0 ng/mL, regardless of risk status.

TABLE 1 Selected characteristics of the pre-intervention and intervention cohorts.

Characteristic	Pre-intervention	Intervention	*p*
Time period	November 11, 2017–February 28, 2019	November 11, 2020–February 28, 2022	—
No. of patients	4022	4252	—
No. of primary care providers	79	84	—
Total no. of visits	14,548	13,781	
Age, years			.901
<50	1736 (43.2)	1817 (42.7)	
50–69	2068 (51.4)	2198 (51.7)	
>69	218 (5.4)	237 (5.6)	
Race, No. (%)			<.001
Black	2450 (60.9)	2422 (57.0)	
Not Black	1572 (39.1)	1830 (43.0)	
No. visits per patient, mean (SD)	3.6 (3.7)	3.2 (2.9)	.061
No. visits per patient, No. (%)			<.0001
1	1287 (32.0)	1358 (31.9)	
2–4	1712 (42.6)	1942 (45.7)	
>4	1023 (25.4)	952 (22.4)	
Visits per clinic site, No. (%)			<.0001
Site 1	5619 (38.6)	5328 (38.7)	
Site 2	3158 (21.7)	2075 (15.1)	
Site 3	1134 (7.8)	1683 (12.2)	
Site 4	1357 (9.3)	1572 (11.4)	
Site 5	1844 (12.7)	1855 (13.5)	
Site 6	1436 (9.9)	1268 (9.2)	
Pre-baseline PSA (<2 years), No. (%)			.310
PSA <4 ng/mL	676 (16.8)	678 (15.9)	
PSA ≥4 ng/mL	64 (1.6)	82 (1.9)	
Pre-baseline biopsy, No. (%)			.212
Benign	18 (0.4)	30 (0.7)	
ASAP or HGPIN	9 (0.2)	9 (0.2)	

Abbreviations: ASAP, atypical small acinar proliferation; HGPIN, high grade prostatic intraepithelial neoplasia; PSA, prostate-specific antigen.

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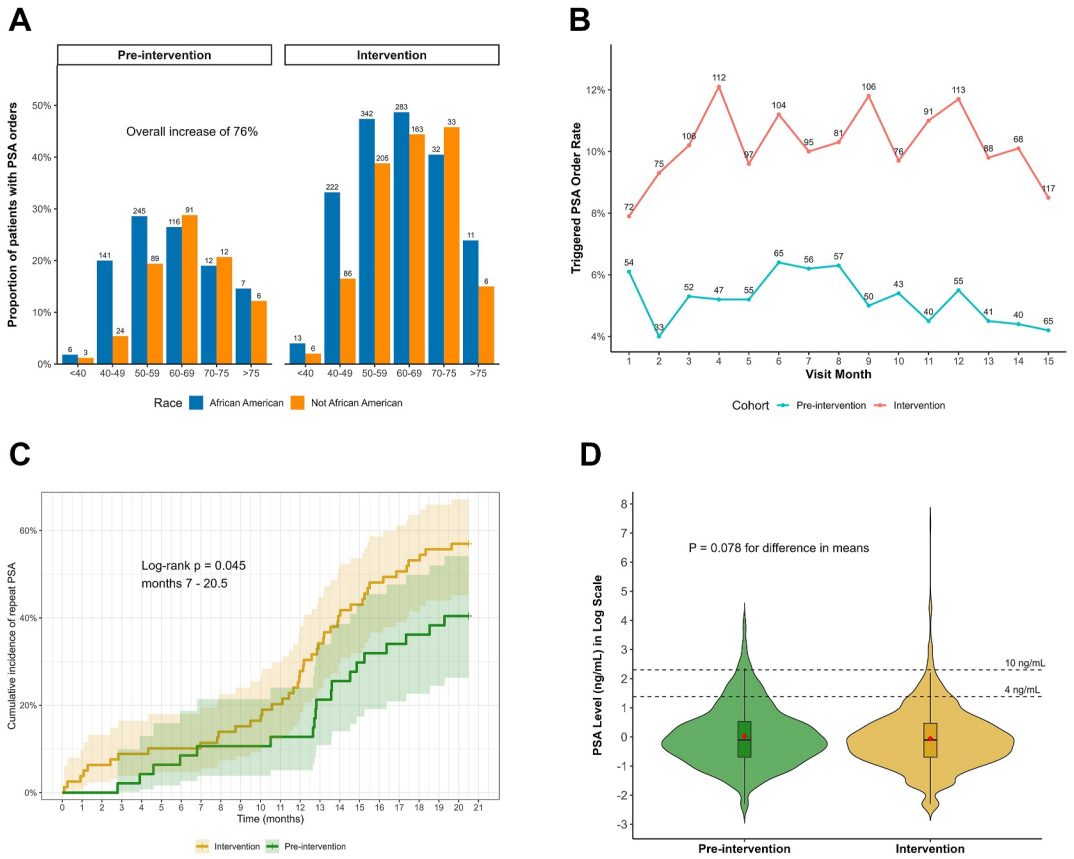


FIGURE 2 (A) Change in percentage of patients screened, stratified by race and age. (B) Percent of eligible patients with triggered PSA orders by month (counts above data points). (C) Cumulative incidence of repeat PSA for patients with PSA 2.5–4.0. (D) Violin plots for distribution of PSA levels and conventional clinical cutoffs at 4 and 10 ng/mL.

Figure 2C shows cumulative incidence curves for repeat PSA from date of the borderline PSA until 20.5 months, the maximum follow-up possible to avoid overlap between periods. The first 6–7 months show a transient increase during intervention, potentially due to confirmatory testing; however, the testing rate thereafter significantly exceeds pre-intervention, reflecting guideline compliance (log-rank $p = .04$). At the end of follow-up, 57.0% versus 40.4% of these patients had repeat tests during the respective periods.

Screening was most likely to occur at the first visit, which also showed the largest difference between intervention and control (Figure S3). However, screening continued to be more frequent during intervention as the total number of visits increased (Table 2). More detailed stratification showed that during pre-intervention, screening increased for patients with two to four visits compared to those with only one visit but then plateaued, whereas during intervention screening continued to increase, plateauing at eight or more ($p < .01$ vs. two to four visits). During both periods, patients with prior PSA either above or below 4.0 had a higher incidence of screening compared to those with no prior PSA; however, these associations were stronger during intervention, when screening increased from 24.2 to 44.3 per 100 among those with prior PSA <4, and from 38.2 to 57.5 among those with prior PSA ≥4. The results were similar for patients with prior benign biopsies—screening increased from 27.8 to 66.7 per 100, and based on small numbers, for men with biopsies showing either ASAP or HGPIN from 33.3 to 77.8.

PSA concentrations within intervals typically considered clinically significant changed very little during intervention (Figure 2D). Mean log PSA was slightly lower for intervention; however, variation in PSA was increased due to more results at both extremes. Biopsy incidence increased during intervention by 60%, from 0.57 to 0.92 biopsies per 100 ($\Delta = 0.35$; 95% CI, −0.03 to 0.75) (Table 3). Cancer prevalence at biopsy increased sharply from 43% to 74% and cumulative incidence of all PCa increased 2.7-fold (0.25 to 0.68 per 100, $p = .007$). Seventy-six percent of PCa was beyond GG1, and incidence beyond GG1 increased approximately 3-fold ($p = .008$). Fifty-two percent of PCa was GG3-5, with a 5-fold increase in incidence relative to the control period. Incidence of cancer with modified NCCN risk level beyond favorable intermediate also increased approximately 3-fold. No pre-intervention biopsies used magnetic resonance imaging (MRI) guidance and among three (of 39) intervention biopsies that did, cancer detection and grading were adequately determined from concurrent 12-core sampling of the gland. Thus, increased use of MRI guidance did not affect biopsy outcomes.

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TABLE 2 Difference between intervention and control period in CumI plus 95% CI stratified by selected subgroups.

	Pre-intervention		Intervention			
	Total *n*	PSA = yes *n* (CumI/100)	Total *n*	PSA = yes *n* (CumI/100)	$CumI_{Int} - CumI_{Pre}$ (95% CI)[a]	p_{diff}[b]
All patients	4022	752 (18.7)	4252	1402 (33.0)	14.3 (12.4–16.1)	<.001
Age group, years, No. (%)						
50–59 (ref.)	1315	334 (25.4)	1250	547 (43.8)	18.4 (14.7–22.0)	Referent
<40	584	9 (1.5)	628	19 (3.0)	1.5 (−0.2 to 3.4)	<.001
40–49	1152	165 (14.3)	1189	308 (25.9)	11.6 (8.3–14.8)	.006
60–69	753	207 (27.5)	948	446 (47.0)	19.6 (14.9–24.1)	.688
70–75	121	24 (19.8)	151	65 (43.0)	23.2 (11.6–33.7)	.419
>75	97	13 (13.4)	86	17 (19.8)	6.4 (−5.2 to 18.2)	.055
Race						
Non-Black all ages	1572	225 (14.3)	1830	499 (27.3)	13.0 (10.2–15.6)	Referent
Black all ages	2450	527 (21.5)	2422	903 (37.3)	15.8 (13.3–18.3)	.136
Non-Black <50 years	664	27 (4.1)	730	92 (12.6)	8.5 (5.6–11.5)	Referent
Black <50 years	898	147 (16.4)	760	235 (30.9)	14.5 (10.4–18.7)	.021
Total visits, No. (%)						
1 (ref.)	1287	109 (8.5)	1358	280 (20.6)	12.2 (9.5–14.8)	Referent
2–4	1712	390 (22.8)	1942	706 (36.4)	13.6 (10.6–16.5)	.488
>4	1023	253 (24.7)	952	416 (43.7)	19.0 (14.8–23.1)	.007
PSA before period						
None (ref.)	3169	535 (16.9)	3100	877 (28.3)	11.4 (9.3–13.5)	Referent
Prior PSA <4	777	188 (24.2)	1039	460 (44.3)	20.1 (15.7–24.3)	<.001
Prior PSA ≥4	76	29 (38.2)	113	65 (57.5)	19.4 (4.0–33.4)	.289
Biopsy before period						
None (ref.)	3995	744 (18.6)	4213	1375 (32.6)	14.0 (12.1–15.9)	Referent
Prior negative biopsy	27	8 (29.6)	39	27 (69.2)	39.6 (12.8–59.7)	.033

Abbreviations: CI, confidence interval; CumI, cumulative incidence of screening per 100 patients; PSA, prostate-specific antigen.

[a]95% CI by modified Wilson method with continuity correction, per Newcombe.[14]

[b]*p* value (two-sided) comparing the difference in CumI versus the difference for the referent subgroup.

DISCUSSION

The primary aim of this trial was to determine the impact of risk-adapted guidelines on PSA screening by care providers in a community health center network serving a majority-Black population with elevated prostate cancer mortality. Although guidelines specifically designed for Black men have been put forward recently, relevant trial-based evidence is scarce.[10] Our findings demonstrate that PCPs can respond to a low-intensity intervention by adopting and sustaining risk-adapted screening, resulting in large increases in screening incidence among eligible patients. The impact was independent of several factors, including patient age, Black race, number of PCP encounters, prior PSA screening, and clinical site, which were associated with screening during both study periods.

Importantly, the intervention effect was not uniform across subgroups, providing evidence for compliance with the guidelines. Screening of Black men under age 50 increased disproportionately and patients with a PSA between 2.5 to 4.0 were more likely to be re-screened as recommended. Screening likelihood continued to increase during intervention even after four encounters, suggesting that PCPs were more persistent in screening these patients despite their greater need for medical attention. Patients with normal PSAs before baseline were more likely to be re tested during intervention, conceivably because many had been primed to accept screening as part of routine care. Patients with elevated PSA or a negative biopsy before baseline were also more likely to be screened, suggesting heightened awareness regarding the need for follow-up. In contrast to these encouraging findings, the data indicate that overscreening of

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TABLE 3 Results of prostate biopsies triggered by PSA orders and visits from Mile Square primary care providers during the pre-intervention versus intervention periods.

	Pre-intervention (n = 23)		Intervention (n = 39)		
	No. (%)	CumI	No. (%)	CumI	p, CumI[a]
A. Pathology diagnosis					
Benign	12 (52)	0.30	7 (18)	0.16	
Isolated HGPIN	1 (4)	0.02	1 (3)	0.02	
Isolated ASAP	0 (0)	0.00	2 (5)	0.05	
Cancer	10 (43)	0.25	29 (74)	0.68	.007
B. All cancers					
GG					
1	3 (30)	0.07	7 (24)	0.16	
2	4 (40)	0.10	7 (24)	0.16	
3	1 (10)	0.02	6 (21)	0.14	
4	1 (10)	0.02	5 (17)	0.12	
5	1 (10)	0.02	4 (14)	0.09	
GG >2	3 (10)	0.07	15 (52)	0.35	.013
GG >1	7 (70)	0.17	22 (76)	0.52	.008
Modified NCCN risk level[b]					
Very low	1 (10)	0.02	3 (10)	0.07	
Low	0 (0)	0.00	2 (7)	0.05	
Favorable intermediate	4 (40)	0.10	7 (24)	0.16	
Unfavorable intermediate	2 (20)	0.05	4 (14)	0.09	
High	1 (10)	0.02	6 (21)	0.14	
Very high	2 (20)	0.05	7 (24)	0.16	
Risk > favorable intermediate	5 (50)	0.12	17 (59)	0.40	.027

Abbreviations: ASAP, atypical small acinar proliferation; GG, grade group; HGPIN, high grade prostatic intraepithelial neoplasia; CumI, cumulative incidence of screening per 100 patient; NCCN, National Comprehensive Cancer Network.

[a]Fisher χ^2 p value, two-sided.

[b]Modified NCCN risk excluding clinical stage and gland volume.

older men might have been exacerbated rather than mitigated by the intervention. This aligns with national data showing increased rates of PSA testing among older men following the 2017 revision of the USPSTF guidelines.[15–17]

As expected, increased screening of younger men shifted PSA levels downward, though this effect was offset by testing of older men with no history of prior screening. The net result was a negligible decrease in the average PSA concentration, and no change in the percentage of patients in clinically significant ranges. However, biopsy incidence rose substantially, and strikingly nearly three out of four biopsies resulted in a cancer diagnosis. In parallel with PSA level changes, screening intensification increased PCa incidence at all risk levels, while having negligible impact on risk-level distribution. During 2016–2019, 46% of all PCa cases diagnosed at UIC had grade higher than GG2 versus 52% in the intervention cohort.

These biopsy results are predictable given a cancer with long pre-clinical latency, a test with long lead-time, and a high-risk population with relatively sparse prior screening exposure. Essentially, given the large pool of latent cases, the intervention, during an initial phase, amplified detection of the same pattern of aggressive cancer that was diagnosed previously. By implication, sustaining the new screening practices beyond a relatively brief 15-month period, will eventually cause a shift toward decreased incidence and lower grade cancer as the pool of latent cases evolves. When that happens is difficult to predict, but meanwhile, the number-needed-to-screen to avoid one death will be reduced because many patients diagnosed with higher-risk cancer will still benefit from earlier access to treatment.[18] Once a new risk equilibrium occurs, it will be critical to modify protocols to reduce detection of clinically nonsignificant PCa. This can include pre-biopsy MRI and deployment of several promising

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biomarkers that can avoid unnecessary diagnosis and treatment.[19,20] Furthermore, the incidence of low-risk PCa, though increased, was still not high, and our previous work in this population showed ready acceptance of active surveillance for indolent tumors.[21]

The updated USPSTF recommendations, moving away from overt discouragement to individualized decisions, reversed a decline in screening, including screening of Black men.[15,22] However, the revised guidelines did not include specific recommendations for men at higher risk.[23,24] The pre-intervention control period was selected so that the USPSTF update, which was publicly announced in April 2017, was mostly "priced in" beforehand. Unlike the case for several other cancers, FQHCs are not required to report their PCa screening activity for quality evaluation purposes, and recent national surveys document substantial uncertainty among PCPs regarding its value.[7,25] Hesitancy about PSA screening is also affected by PCP concerns regarding the feasibility of incorporating shared decision-making due to time pressure, and concerns about implementation for patients with high comorbidity and/or low health literacy.[7]

The evidentiary basis for our risk-adapted guidelines is quite strong.[24] Age-specific curves from SEER for PCa incidence and mortality show higher risks for younger Black men compared to age peers.[10] Furthermore, the value of baseline PSA for evaluating risk at mid-life is now firmly established. Prospective analyses from large cohorts demonstrated that baseline levels of PSA at relatively young ages are stronger predictors of lethal PCa than either race or family history.[13,26–29] The salience of mid-life PSA was also confirmed in a cohort analysis of Black men recruited from community health centers.[30] Although recent model-based recommendations for Black men favor starting screening at age 45, we adopted age 40 because clinic appearance can be unpredictable for men under age 50.[31]

Certain limitations pertinent to this study should be noted. The intervention overlapped with the COVID pandemic, which created additional challenges for PCPs. It was encouraging that the number of unique patients and total visits were relatively unchanged. However, 17.6% of patient encounters during intervention occurred via telehealth, which was associated with a lower screening activity than in-person visits and suggests that the intervention impact could be underestimated. Although attention to family history was included in the intervention, we were unable to address it in analyses because data quality in the medical records was deemed inadequate. The shift in electronic health record software from Cerner to Epic between study periods meant that some variables of interest, such as visit type, could not be harmonized and thus compared. Unfortunately, we were unable to collect data to understand whether PSA orders on men over age 69 were due to patient insistence, inaccurate estimation of life expectancy, or heightened PCP concerns about prostate-related symptoms.

Additional studies are needed to examine provider-patient interaction in finer detail than was possible in this study. It appears plausible that continuity of care, provider characteristics, and level of trust experienced by both provider and patient, could significantly affect adoption of risk-adapted PCa screening.[32] In addition, validation of better tools for PCPs to assess life expectancy and greater emphasis in PCP training could reduce overscreening of older men.[33]

To our knowledge, this is the first trial to demonstrate that PCPs practicing in high-risk community settings can adopt and sustain a simple protocol for risk-adapted PSA screening. Although such protocols could eventually reduce the incidence of potentially lethal PCa, there will be a transition period during which this incidence will actually increase. Our results are consistent with a recent model for decomposing the racial disparity in PCa mortality, which found that the burden of preclinical disease and the higher prevalence of aggressive features at diagnosis among Black men support the need to deploy better methods for early detection.[6] Indeed, another modeling analysis concluded that with long-term follow-up, PSA screening has substantially greater net benefit for Black men relative to the general population.[18]

We do not consider our Smart PSA guidelines to be definitive; trials of other guidelines with intermediate end points are needed, including trials with randomized cluster designs, because trials with long-term end points have pragmatic limitations. In the near term, designs that rely on expensive or inaccessible tests might not be feasible in under-resourced settings.[34,35] The ongoing PROBASE trial in Germany is testing men starting at age 45–50 using baseline PSA-stratified repeat testing intervals.[36] Early results are becoming available but must be translated with caution to populations with different risk profiles and health care conditions.[37] The TRANSFORM trial recently launched in the United Kingdom seeks to identify optimal risk-adapted protocols using MRI and polygenic risk scores in addition to PSA. The organizers are making a concerted effort to enroll men from high-risk, vulnerable groups; however, final results are not expected for at least a decade.[38] We focused the intervention on PCPs because they are responsible for most cancer screening, and because interventions during episodic care can be quite efficient.[39] Once PCPs are activated to screen, however, important opportunities will emerge to generate synergy by activating patients through coordinated community outreach.

AUTHOR CONTRIBUTIONS

Peter H. Gann: Conceptualization, visualization, data curation, formal analysis, methodology, writing—original draft, supervision, and project administration. **Nathan Stackhouse**: Data curation, writing—review and editing, supervision, and project administration. **Nicole Gastala**: Data curation, writing—review and editing, supervision, and project administration. **Weiwei Ma**: Data curation, formal analysis, and writing—review and editing. **Margaret E. Wright**: Conceptualization, data curation, and writing—review and editing. **Karriem Watson**: Conceptualization, writing—review and editing, and project administration. **Christopher Stepping**: Data curation and writing—review and editing. **Patrice King-Lee**: Data curation, writing—review and editing, supervision, and project administration. **Ziqiao Xu**: Data curation, formal analysis, and writing—review and editing. **Tushar Patel**: Data curation and writing—review and editing. **Michael R. Abern**: Conceptualization, data curation, methodology, and writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

Christopher Stepping reports fees for professional activities from the University of Illinois at Chicago. Ziqiao Xu reports fees for professional activities from George Mason University. The other authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data underlying this article were provided by the Mile Square Health Center and the University of Illinois Health system (UI Health). De-identified data used for the study can be shared on request to the corresponding author with the permission of both of these entities.

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